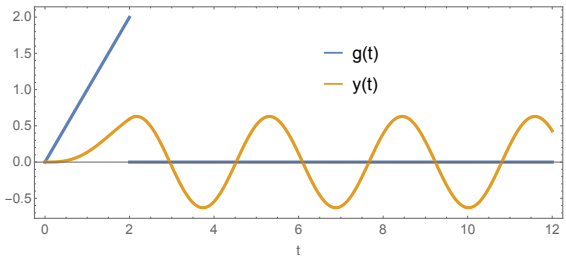
Lecture 13 recap

In Lecture 13:

- Went through examples on using LTs to solve IVPs with discontinuous forcing
  - Looked at off–on forcing and how this produces discontinuities in a solution derivative
  - Learnt how to represent a ramp-like forcing using Heaviside step functions

$$y(t) = \frac{1}{4}t - \frac{1}{8}\sin 2t - \left(\frac{t}{4} - \frac{1}{2}\cos 2(t-2) - \frac{1}{8}\sin 2(t-2)\right) H(t-2)$$



Lecture 14

In Lecture 14, we will:

1. Introduce the Dirac delta and use it to model impulsive forcing
2. Introduce convolutions as a tool to calculate ILTs

Impulses

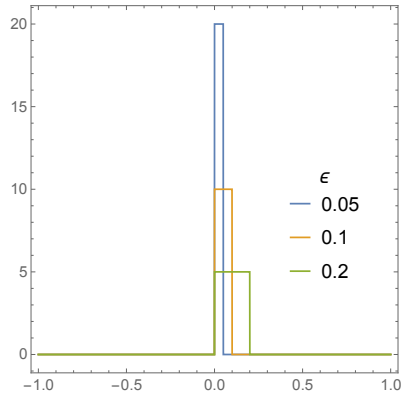
Suppose we want to model a force acting over a negligibly small time period, i.e., an impulse

Use a rectangular pulse of unit area:

$$\delta_\epsilon(t) = \begin{cases} 0 & t \leqslant 0 \\ \frac{1}{\epsilon} & 0 < t \leqslant \epsilon \\ 0 & t > \epsilon \end{cases}, \quad \text{“small” } \epsilon > 0$$

The ‘strength’ of this impulse is

$$\int_0^\infty \delta_\epsilon(t)\,\mathrm{d}t = \int_0^\epsilon \frac{1}{\epsilon}\,\mathrm{d}t = 1$$



As width $\to 0$, height $\to \infty$
but area remains 1

Limit as $\epsilon \to 0$

1. $\lim_{\epsilon \to 0} \delta_\epsilon = 0$ for all $t \neq 0$
2. $\lim_{\epsilon \to 0} \int_0^\infty \delta_\epsilon \, \mathrm{d}t = 1$
3. $\lim_{\epsilon \to 0} \mathscr{L}(\delta_\epsilon) = \lim_{\epsilon \to 0} \dfrac{1 - \mathrm{e}^{-\epsilon s}}{\epsilon s} = 1$ (L'Hôpital's rule)

No function (in the traditional sense) can have these properties!

The Dirac delta, $\delta(t)$

Definition

We define $\delta(t)$ such that

1. $\delta(t) = 0$ for all $t \neq 0$,
2. $\displaystyle\int_{-\infty}^{\infty} \delta(t)\mathrm{d}t = \int_{0}^{a} \delta(t)\mathrm{d}t = 1$ for all $a > 0$, and
3. $\mathscr{L}\{\delta(t)\} = 1$.

- $\delta(t)$ is not a function in the traditional sense (it is a *generalised function* as you will discover in later courses)
- There are various forms of $\delta(t)$ with others being centred at or left of zero; we will always use the above definition

Sifting property

- Since $\delta(t)$ is zero for $t \neq 0$, it 'sifts out' the value of a function at $t = 0$

$$\int_0^\infty f(t)\delta(t)\mathrm{d}t = f(0)$$

- Let $c \geqslant 0$, then

1. $\displaystyle\int_0^\infty f(t)\delta(t-c)\ \mathrm{d}t = f(c)$
2. $\mathscr{L}\Big\{\delta(t-c)\Big\} = \displaystyle\int_0^\infty \mathrm{e}^{-st}\delta(t-c)\ \mathrm{d}t = \mathrm{e}^{-sc}$

Exercise 3.19: Solve $y'(t) - y(t) = \delta(t-1)$ where $y(0) = 0$

We solve the problem using Laplace transforms.

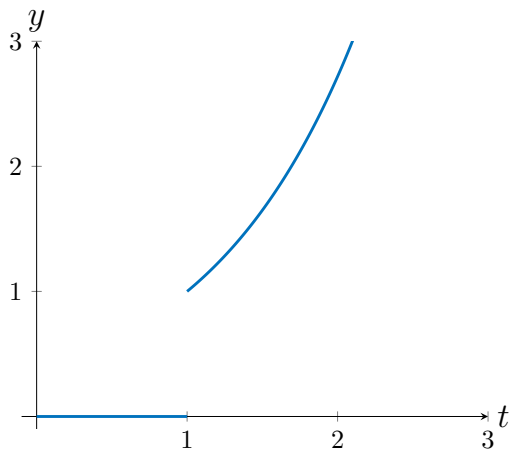
$$\mathscr{L}\{y' - y\} = \mathscr{L}\{\delta(t-1)\}$$

$$\Rightarrow \quad sY - Y = \mathrm{e}^{-s}$$

$$Y(s) = \frac{\mathrm{e}^{-s}}{s-1}$$

$$y(t) = \mathscr{L}^{-1}\left(\frac{\mathrm{e}^{-s}}{s-1}\right) = \mathrm{e}^{t-1}H(t-1) \quad t\text{-shifting}$$

$$y(t) = \mathrm{e}^{t-1} H(t-1)$$



The solution satisfies the initial condition and the ODE but it is discontinuous

Convolution

Definition (Convolution)

Given two functions, f and g, defined on $[0, \infty)$, their convolution, $f * g$ is defined

$$f * g = \int_0^t f(t - \tau) g(\tau) \mathrm{d}\tau$$

Basic properties

- Convolutions are commutative, i.e., $f * g = g * f$
- In general, $1 * f \neq f$, as

$$(1 * f)(t) = \int_0^t f(\tau) \mathrm{d}\tau, \quad \text{i.e., an antiderivative}$$

Exercise 3.20: Calculate the convolution of e^t with e^{2t}

$$\begin{aligned} \mathrm{e}^t * \mathrm{e}^{2t} &= \int_0^t \mathrm{e}^{t-\tau}\mathrm{e}^{2\tau}\ \mathrm{d}\tau \\ &= \mathrm{e}^t \int_0^t \mathrm{e}^{\tau}\mathrm{d}\tau \\ &= \mathrm{e}^t(\mathrm{e}^t - 1) \end{aligned}$$

Note that

$$\mathrm{e}^{2t} * \mathrm{e}^t = \int_0^t \mathrm{e}^{2(t-\tau)}\mathrm{e}^{\tau}\ \mathrm{d}\tau = \mathrm{e}^{2t}\int_0^t \mathrm{e}^{-\tau}\ \mathrm{d}\tau = \mathrm{e}^{2t}(1-\mathrm{e}^{-t}) = \mathrm{e}^t(\mathrm{e}^t-1)$$

Exercise 3.21: Calculate $\mathscr{L}^{-1}\left\{\dfrac{1}{(s-1)(s-2)}\right\}$

Laplace transform of the convolution

Let f and g be functions with Laplace transforms, then

$$\mathscr{L}(f * g) = F(s)G(s)$$

Laplace transforms convert convolutions into multiplication

$$\mathscr{L}^{-1}\left\{\frac{1}{s-1}\right\} = \mathrm{e}^{t} \quad \text{and} \quad \mathscr{L}^{-1}\left\{\frac{1}{s-2}\right\} = \mathrm{e}^{2t}$$

$$\Rightarrow \quad \mathscr{L}^{-1}\left\{\frac{1}{s-1}\frac{1}{s-2}\right\} = \mathrm{e}^{t} * \mathrm{e}^{2t}$$

From previous exercise $\mathrm{e}^{t} * \mathrm{e}^{2t} = \mathrm{e}^{t}(\mathrm{e}^{t} - 1)$

Exercise 3.22: Solve the following integro-differential equation

$$y'(t) + 4\int_0^t y(\tau)\mathrm{d}\tau = 0 \qquad \text{where} \qquad y(0) = 3$$

Laplace transform of an antiderivative

$$\mathscr{L}\left(\int_0^t f(\tau)\mathrm{d}\tau\right) = \mathscr{L}(1 * f) = \mathscr{L}(1)F(s) = \frac{F(s)}{s}$$

Taking the Laplace transform gives

$$\begin{aligned} sY - 3 + 4Y * \mathscr{L}(1) = 0 \quad &\Rightarrow \quad sY - 3 + \frac{4Y}{s} = 0 \\ s^2Y - 3s + 4Y = 0 \quad &\Rightarrow \quad (s^2+4)Y = 3s \\ Y = \frac{3s}{s^2+4} \quad &\Rightarrow \quad Y = \mathscr{L}^{-1}\left(\frac{3s}{s^2+4}\right) = 3\cos 2t \end{aligned}$$